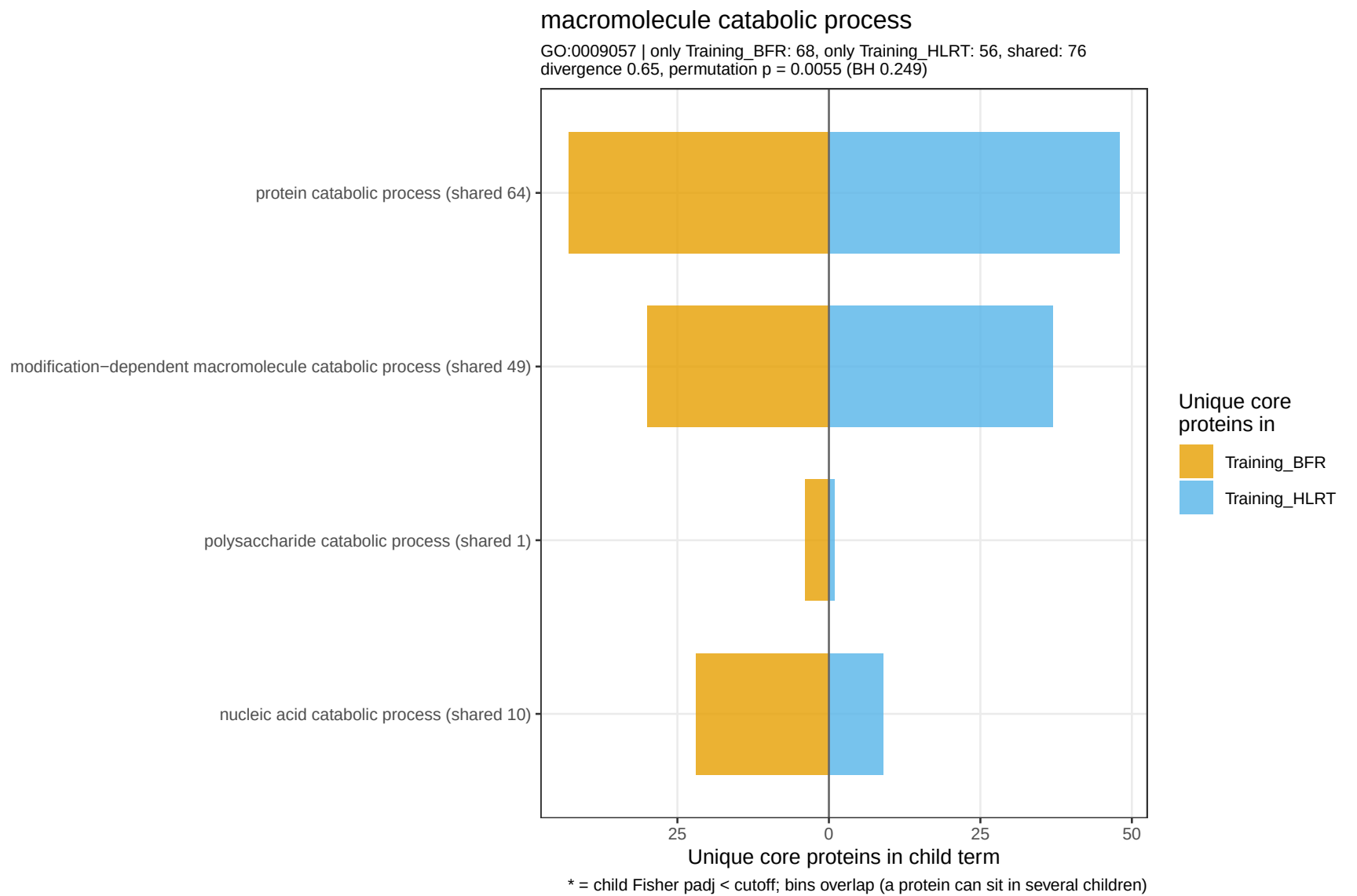
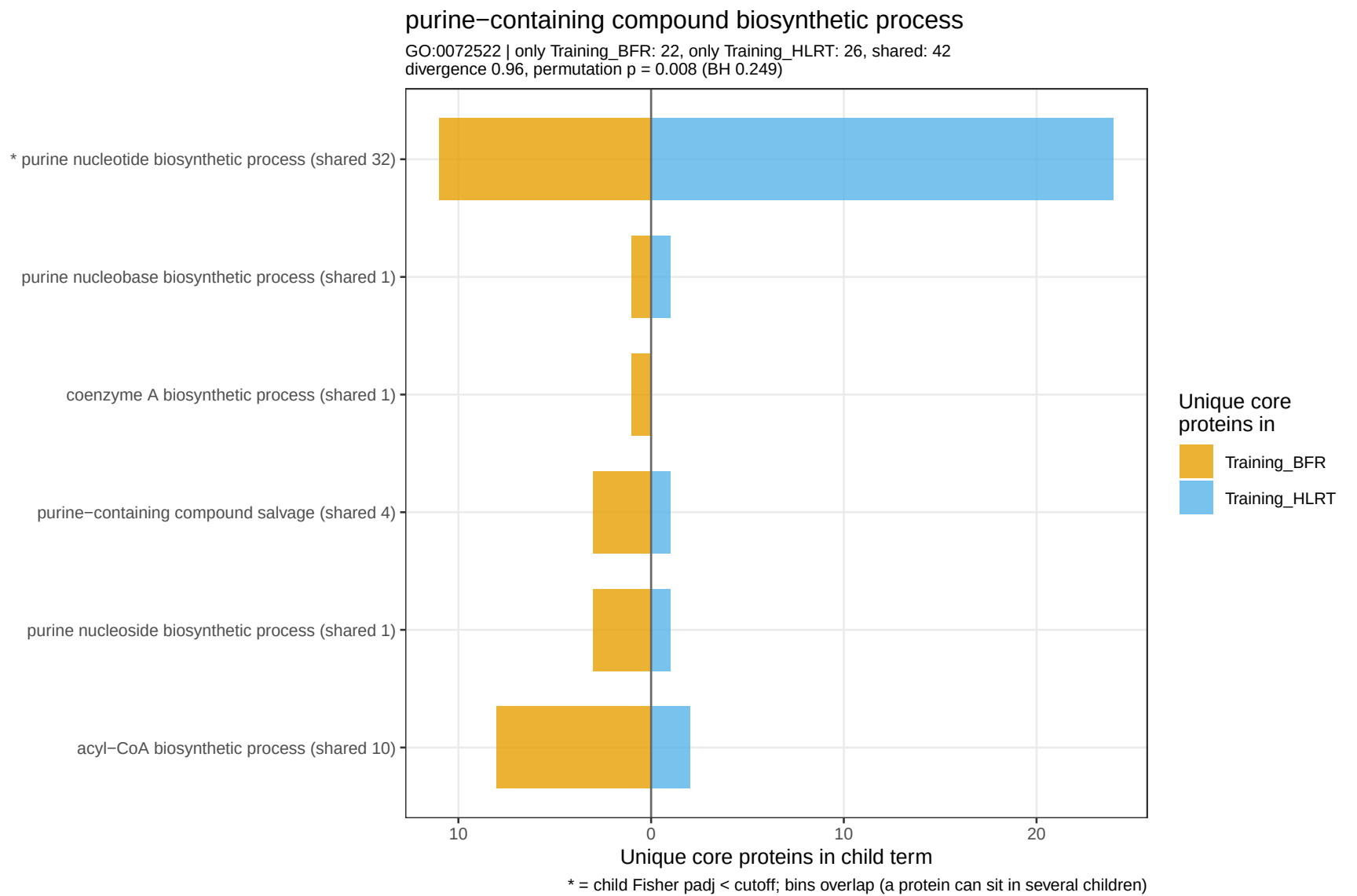
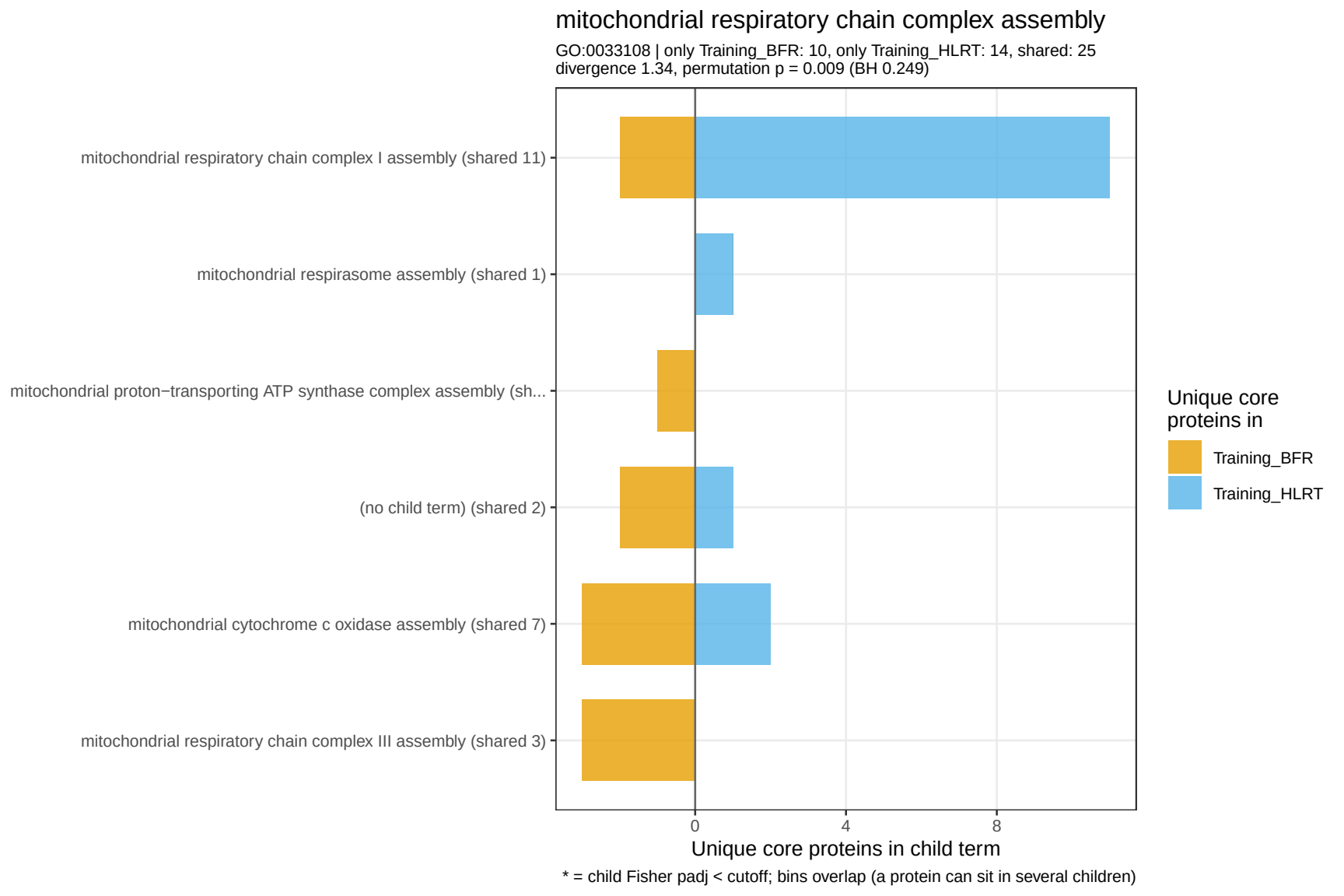
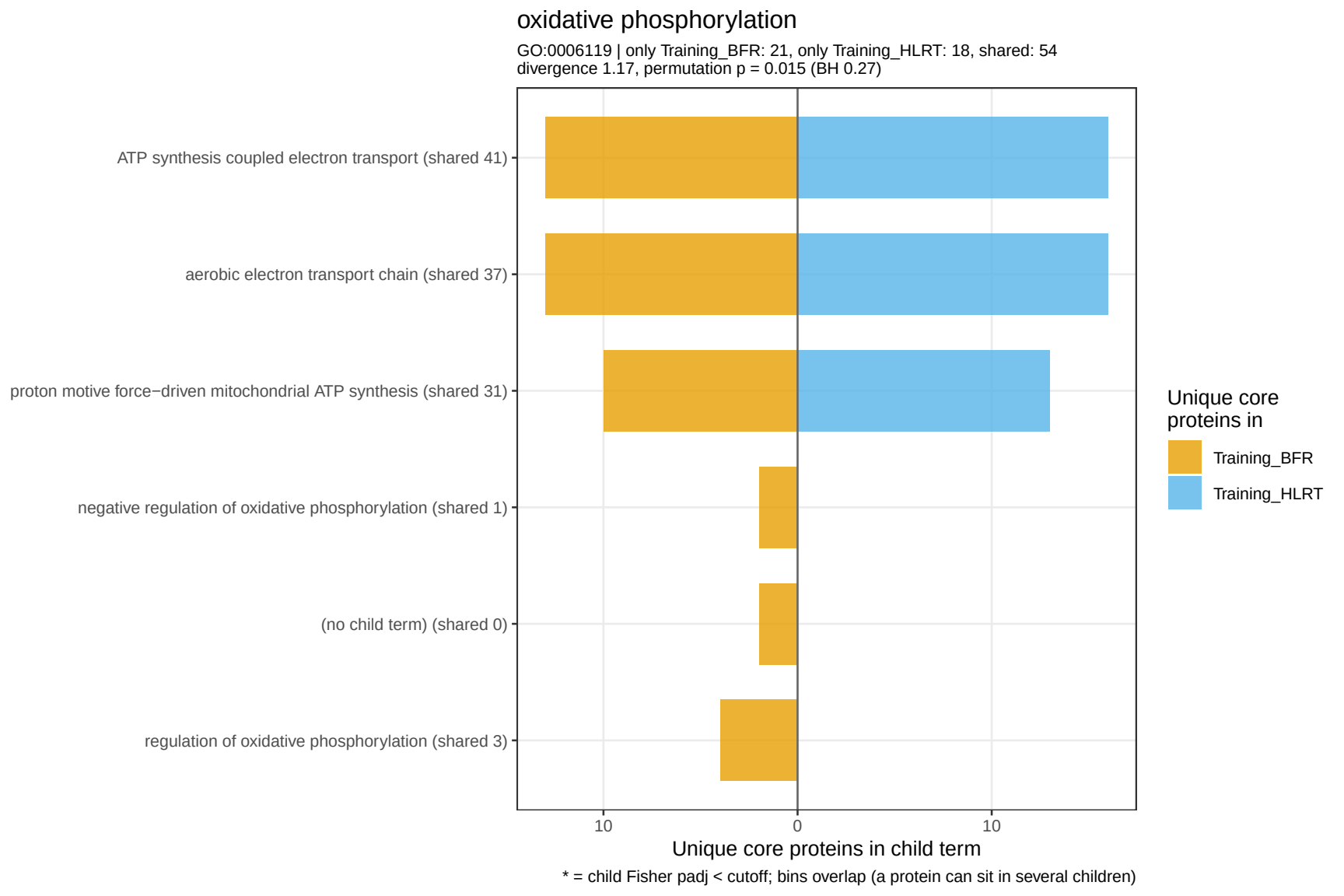
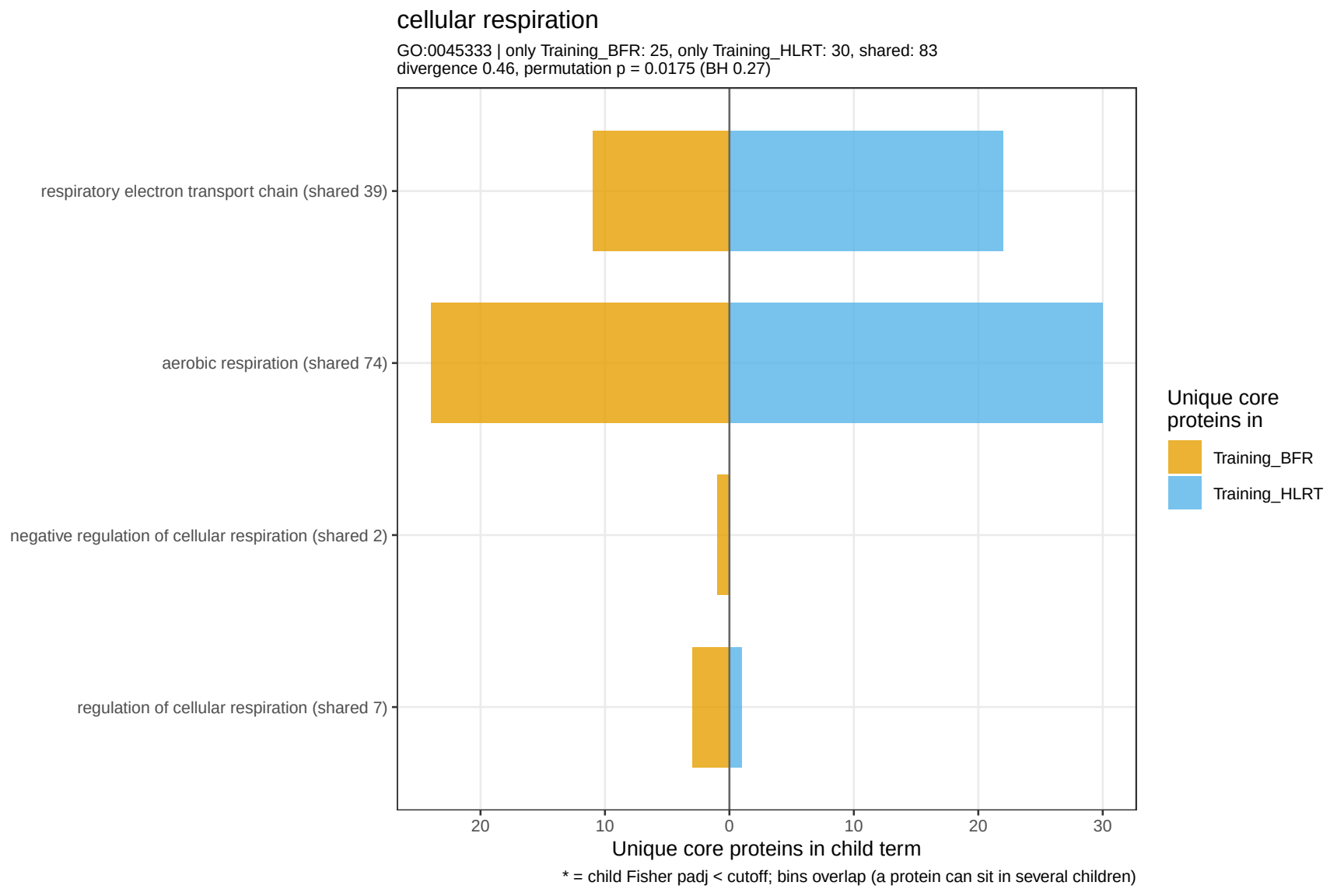
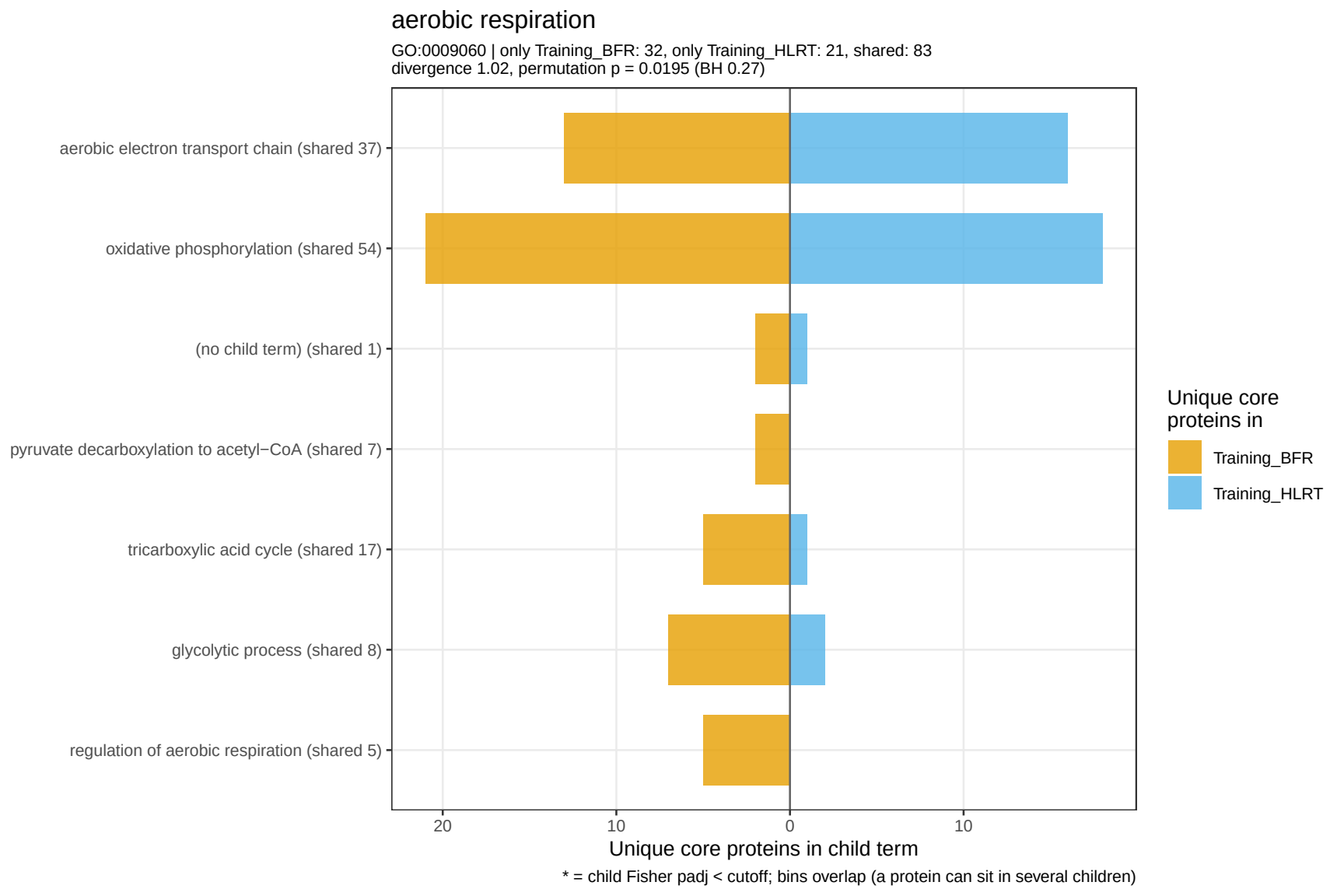

macromolecule catabolic process
GO:0009057 | only Training_BFR: 68, only Training_HLRT: 56, shared: 76
divergence 0.65, permutation p = 0.0055 (BH 0.249)
protein catabolic process (shared 64)
modification−dependent macromolecule catabolic process (shared 49)
polysaccharide catabolic process (shared 1)
nucleic acid catabolic process (shared 10)
25
0
25
50
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

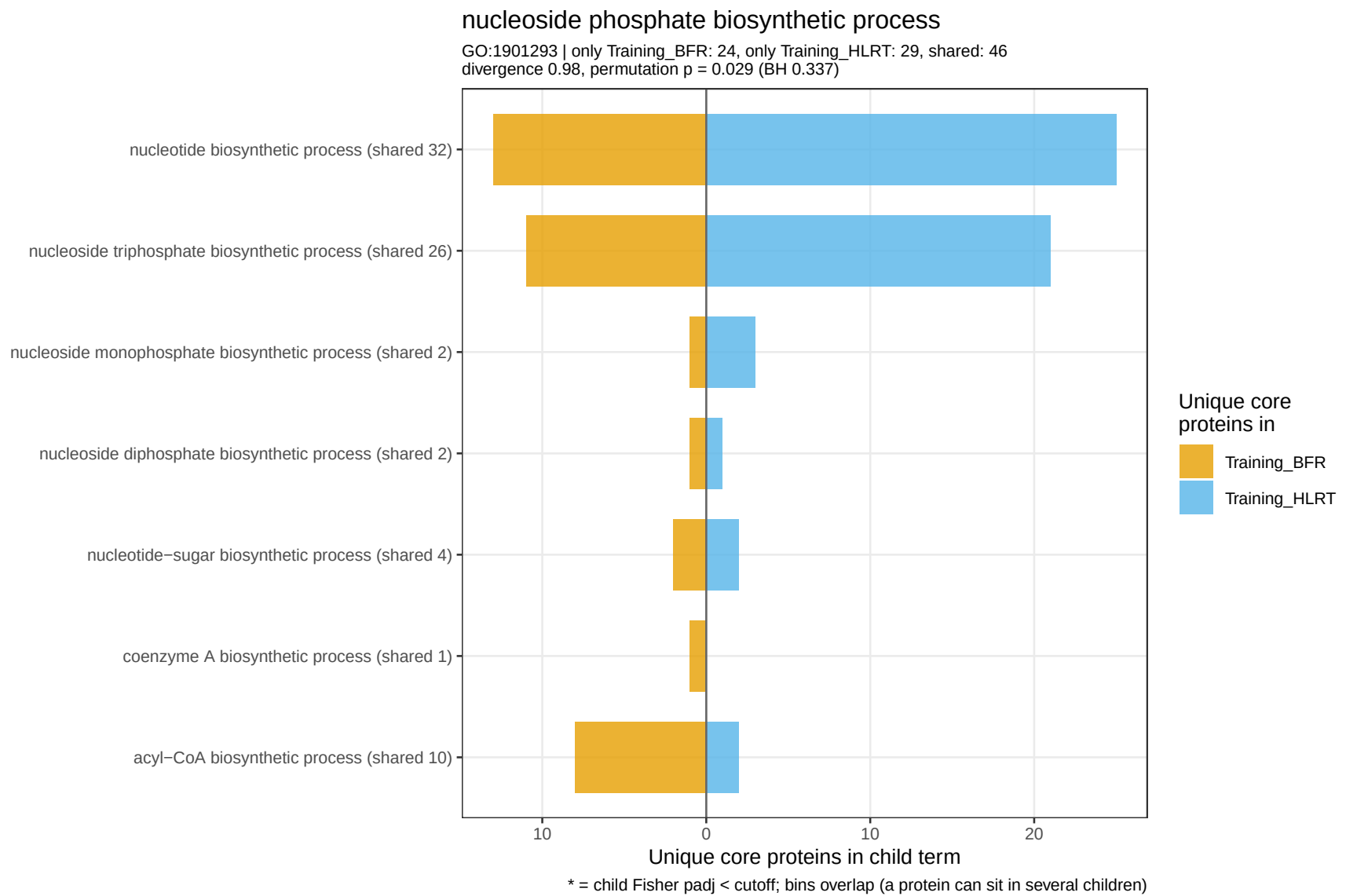

purine−containing compound biosynthetic process
GO:0072522 | only Training_BFR: 22, only Training_HLRT: 26, shared: 42
divergence 0.96, permutation p = 0.008 (BH 0.249)
* purine nucleotide biosynthetic process (shared 32)
purine nucleobase biosynthetic process (shared 1)
coenzyme A biosynthetic process (shared 1)
purine−containing compound salvage (shared 4)
purine nucleoside biosynthetic process (shared 1)
acyl−CoA biosynthetic process (shared 10)
10
0
10
20
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

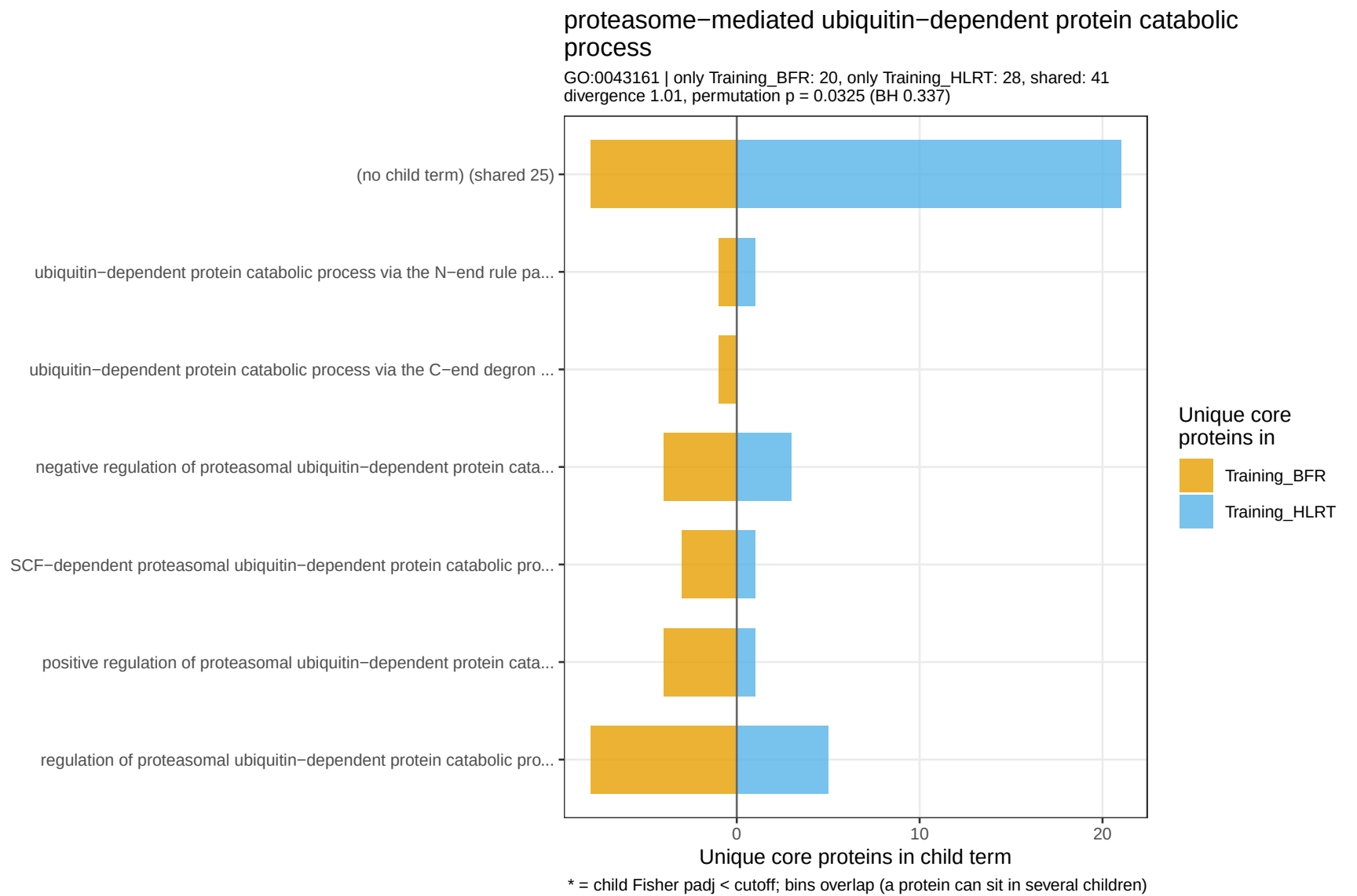

mitochondrial respiratory chain complex assembly
GO:0033108 | only Training_BFR: 10, only Training_HLRT: 14, shared: 25
divergence 1.34, permutation p = 0.009 (BH 0.249)
mitochondrial respiratory chain complex I assembly (shared 11)
mitochondrial respirasome assembly (shared 1)
mitochondrial proton−transporting ATP synthase complex assembly (sh...
(no child term) (shared 2)
mitochondrial cytochrome c oxidase assembly (shared 7)
mitochondrial respiratory chain complex III assembly (shared 3)
0
4
8
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

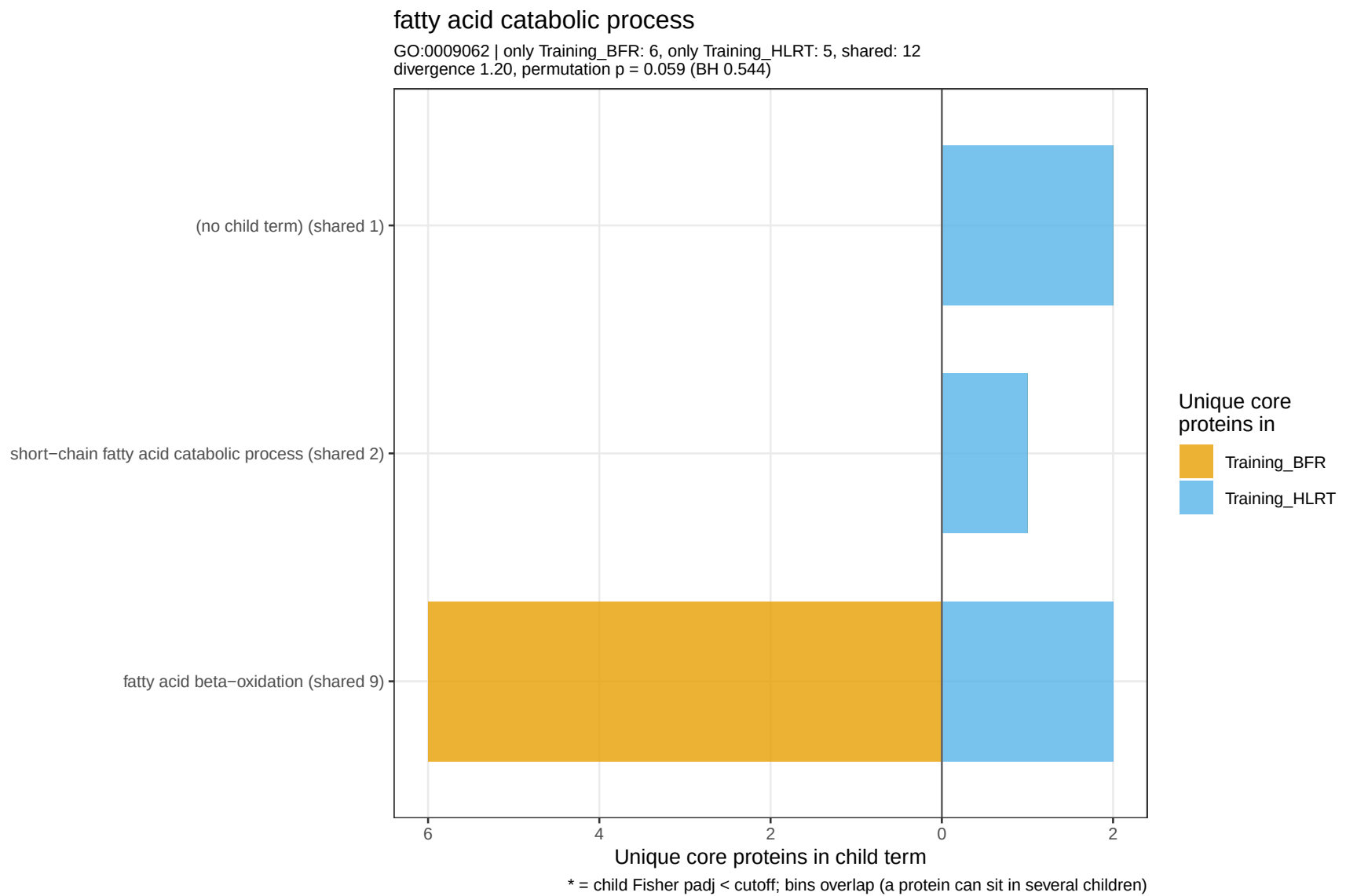

oxidative phosphorylation
GO:0006119 | only Training_BFR: 21, only Training_HLRT: 18, shared: 54
divergence 1.17, permutation p = 0.015 (BH 0.27)
ATP synthesis coupled electron transport (shared 41)
aerobic electron transport chain (shared 37)
proton motive force−driven mitochondrial ATP synthesis (shared 31)
negative regulation of oxidative phosphorylation (shared 1)
(no child term) (shared 0)
regulation of oxidative phosphorylation (shared 3)
10
0
10
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

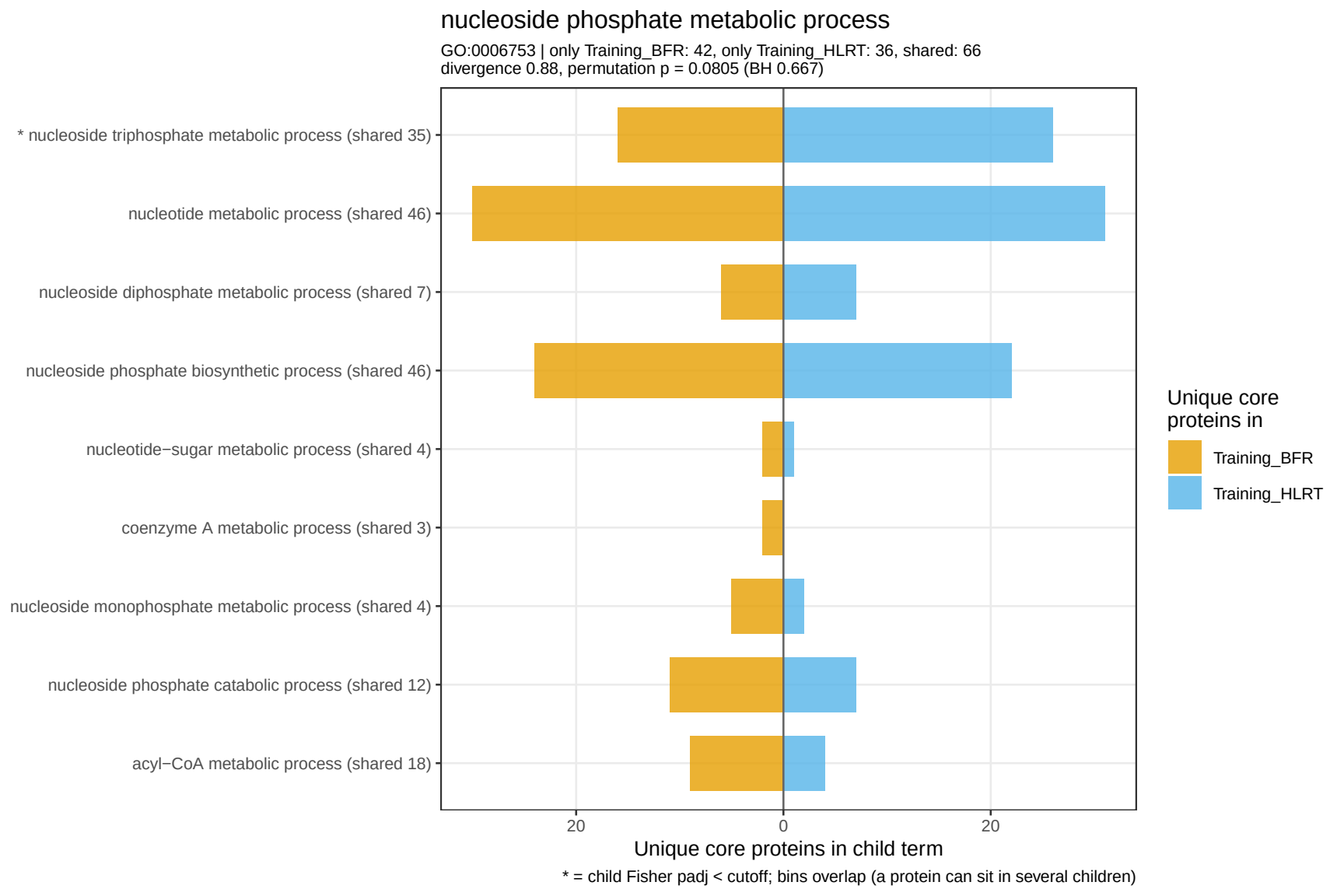

cellular respiration
GO:0045333 | only Training_BFR: 25, only Training_HLRT: 30, shared: 83
divergence 0.46, permutation p = 0.0175 (BH 0.27)
respiratory electron transport chain (shared 39)
aerobic respiration (shared 74)
negative regulation of cellular respiration (shared 2)
regulation of cellular respiration (shared 7)
20
10
0
10
20
30
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

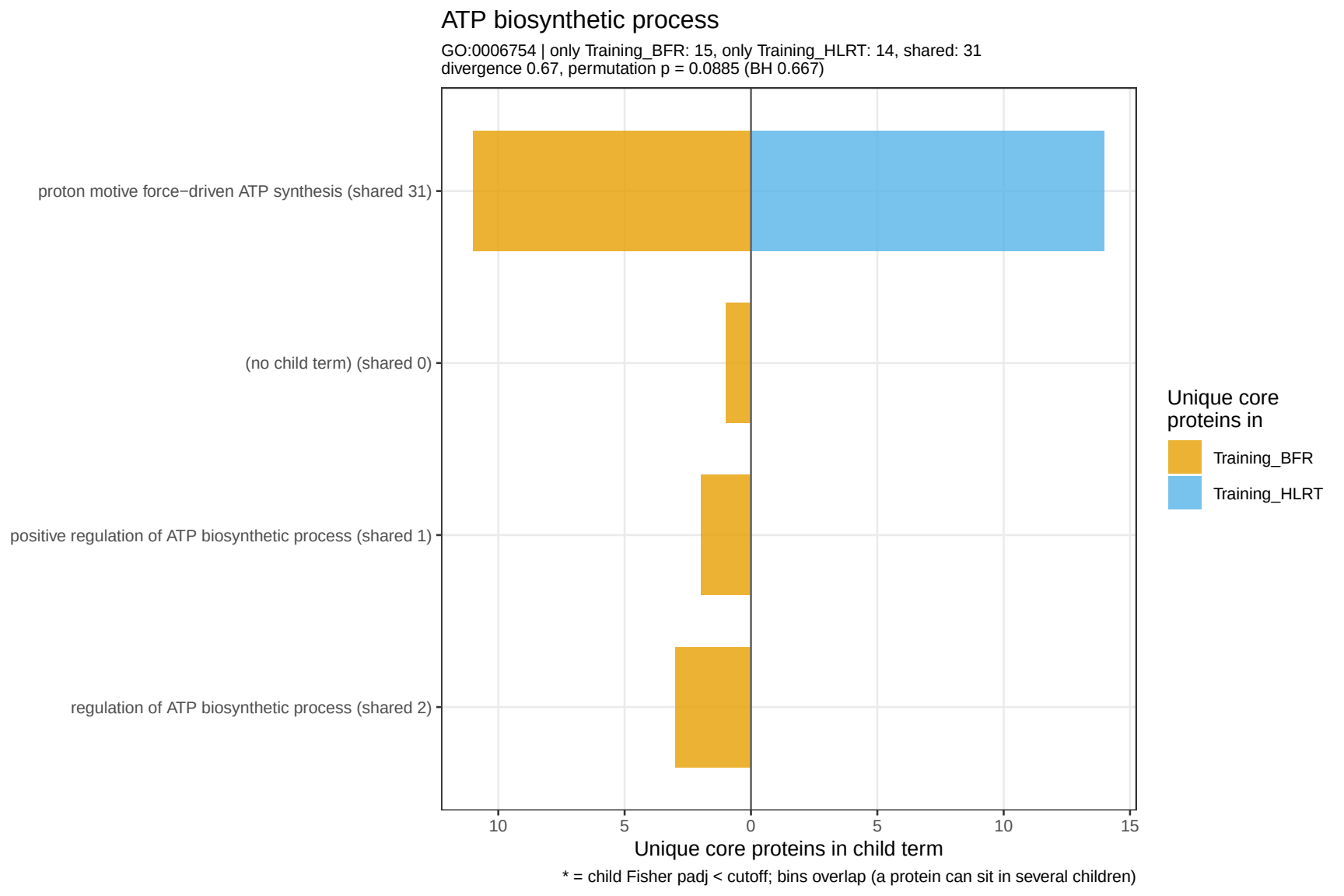

aerobic respiration
GO:0009060 | only Training_BFR: 32, only Training_HLRT: 21, shared: 83
divergence 1.02, permutation p = 0.0195 (BH 0.27)
aerobic electron transport chain (shared 37)
oxidative phosphorylation (shared 54)
(no child term) (shared 1)
pyruvate decarboxylation to acetyl−CoA (shared 7)
tricarboxylic acid cycle (shared 17)
glycolytic process (shared 8)
regulation of aerobic respiration (shared 5)
20
10
0
10
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

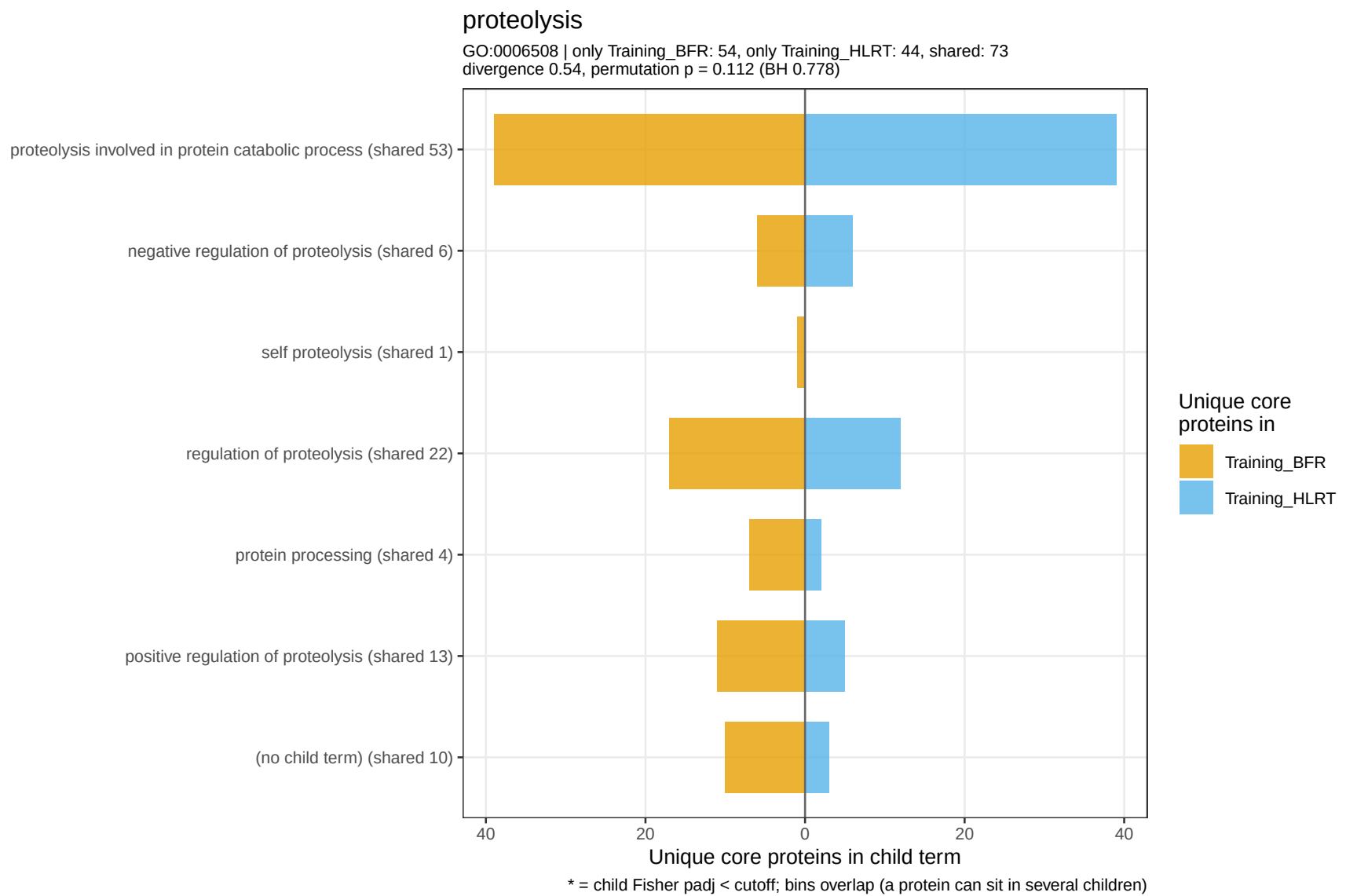

nucleoside phosphate biosynthetic process
GO:1901293 | only Training_BFR: 24, only Training_HLRT: 29, shared: 46
divergence 0.98, permutation p = 0.029 (BH 0.337)
nucleotide biosynthetic process (shared 32)
nucleoside triphosphate biosynthetic process (shared 26)
nucleoside monophosphate biosynthetic process (shared 2)
nucleoside diphosphate biosynthetic process (shared 2)
nucleotide−sugar biosynthetic process (shared 4)
coenzyme A biosynthetic process (shared 1)
acyl−CoA biosynthetic process (shared 10)
10
0
10
20
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

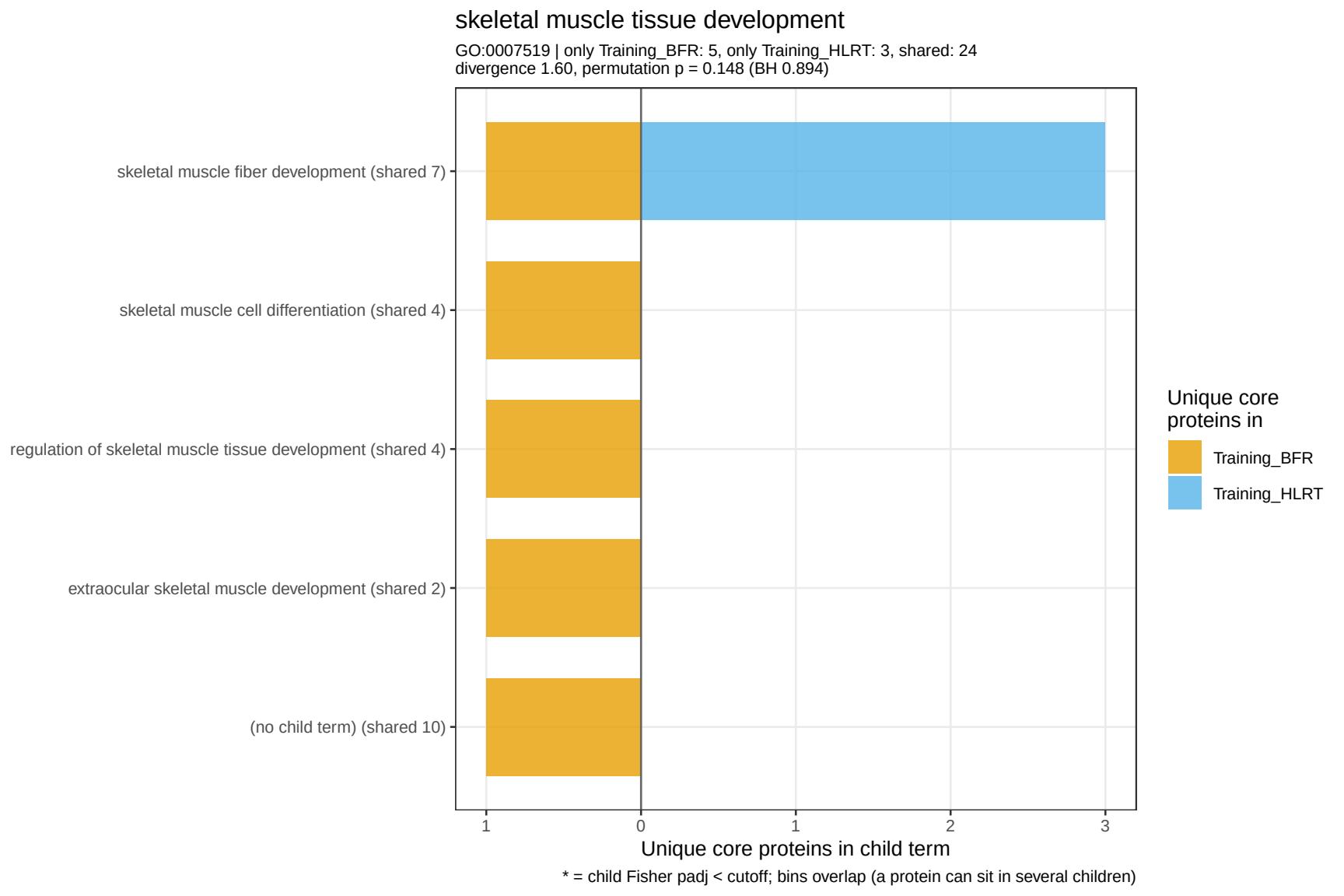

proteasome−mediated ubiquitin−dependent protein catabolic process
GO:0043161 | only Training_BFR: 20, only Training_HLRT: 28, shared: 41
divergence 1.01, permutation p = 0.0325 (BH 0.337)
(no child term) (shared 25)
ubiquitin−dependent protein catabolic process via the N−end rule pa...
ubiquitin−dependent protein catabolic process via the C−end degron ...
negative regulation of proteasomal ubiquitin−dependent protein cata...
SCF−dependent proteasomal ubiquitin−dependent protein catabolic pro...
positive regulation of proteasomal ubiquitin−dependent protein cata...
regulation of proteasomal ubiquitin−dependent protein catabolic pro...
0
10
20
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

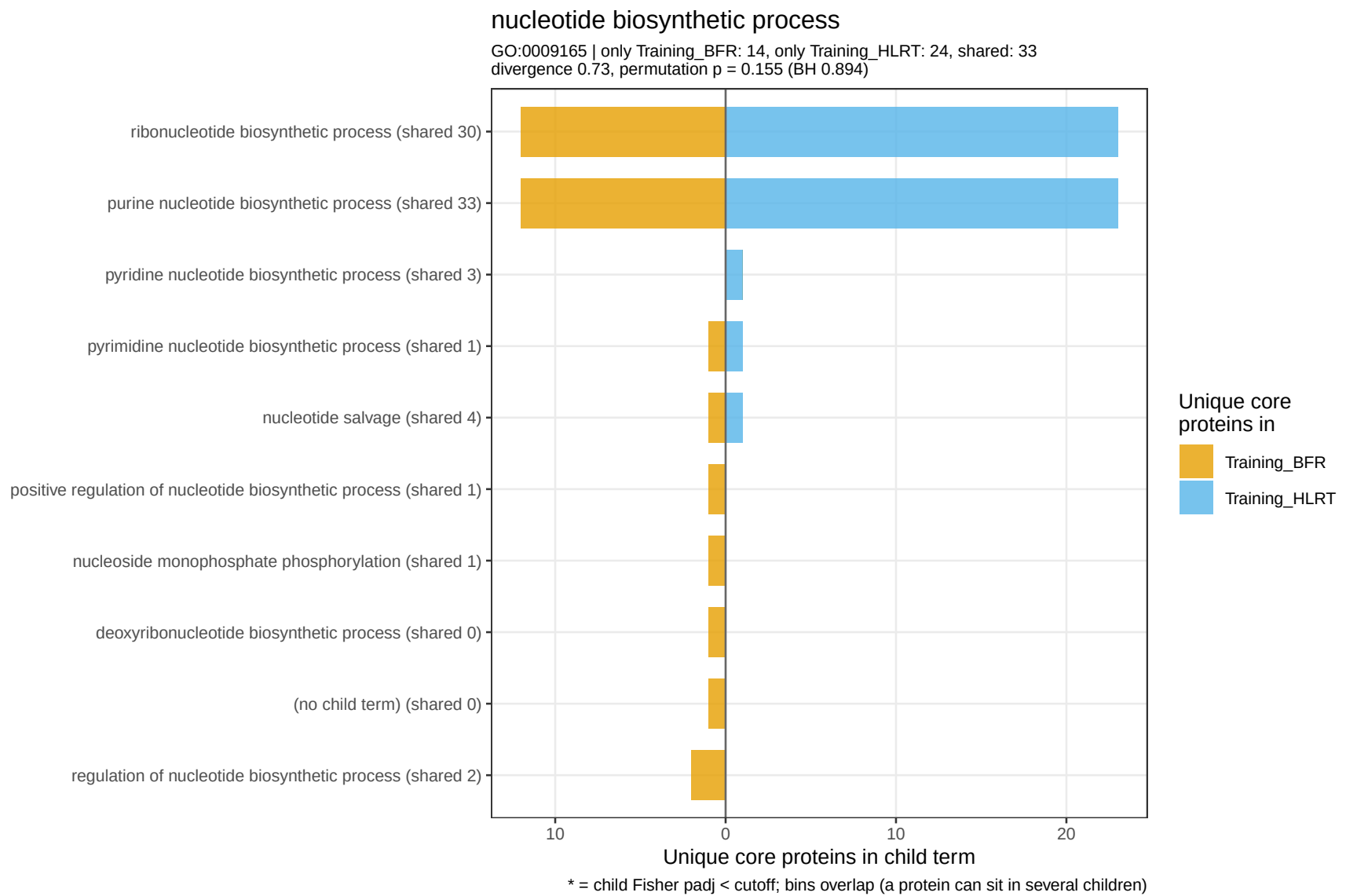

fatty acid catabolic process
GO:0009062 | only Training_BFR: 6, only Training_HLRT: 5, shared: 12
divergence 1.20, permutation p = 0.059 (BH 0.544)
(no child term) (shared 1)
short−chain fatty acid catabolic process (shared 2)
fatty acid beta−oxidation (shared 9)
6
4
2
0
2
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

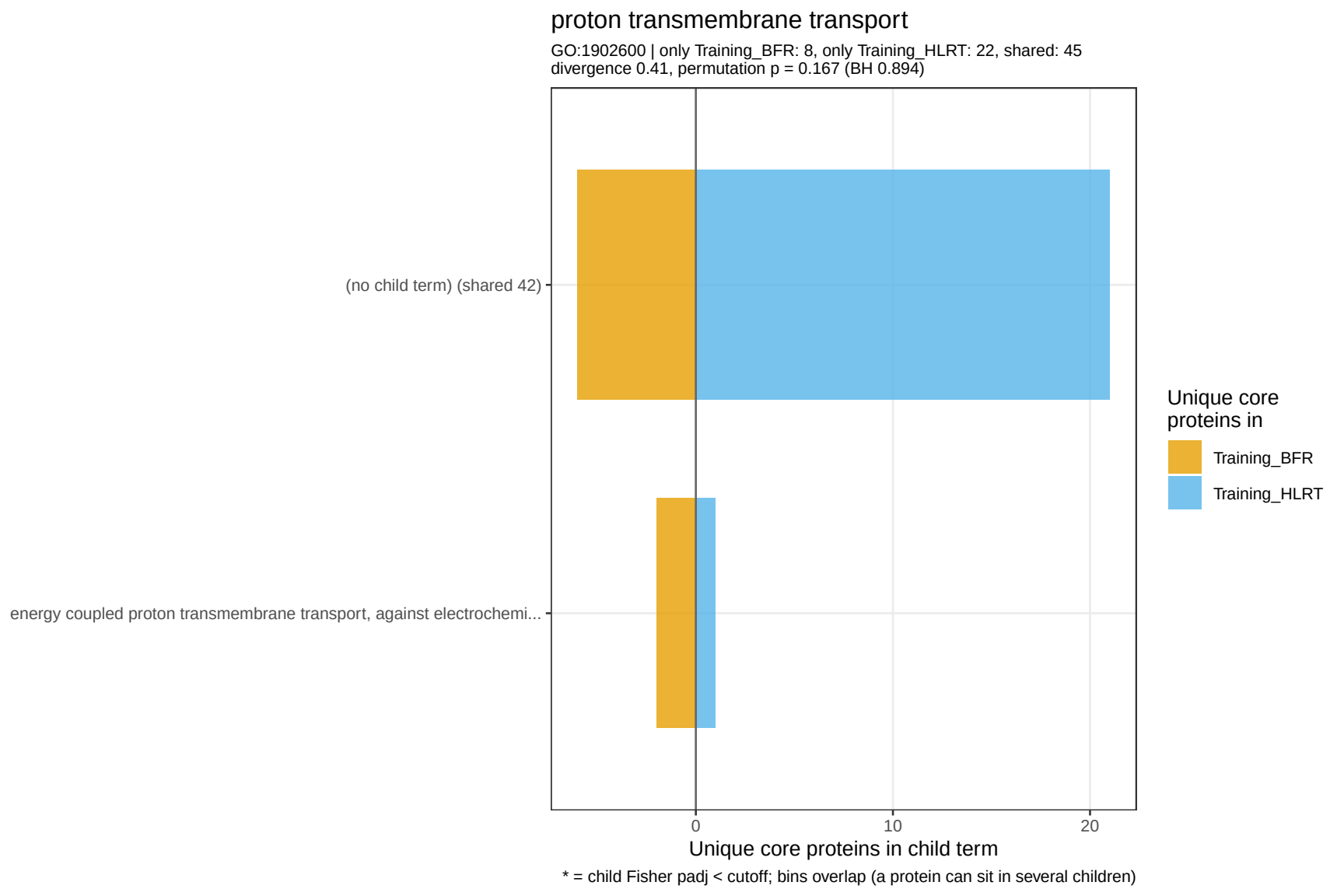

nucleoside phosphate metabolic process
GO:0006753 | only Training_BFR: 42, only Training_HLRT: 36, shared: 66
divergence 0.88, permutation p = 0.0805 (BH 0.667)
* nucleoside triphosphate metabolic process (shared 35)
nucleotide metabolic process (shared 46)
nucleoside diphosphate metabolic process (shared 7)
nucleoside phosphate biosynthetic process (shared 46)
nucleotide−sugar metabolic process (shared 4)
coenzyme A metabolic process (shared 3)
nucleoside monophosphate metabolic process (shared 4)
nucleoside phosphate catabolic process (shared 12)
acyl−CoA metabolic process (shared 18)
20
0
20
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

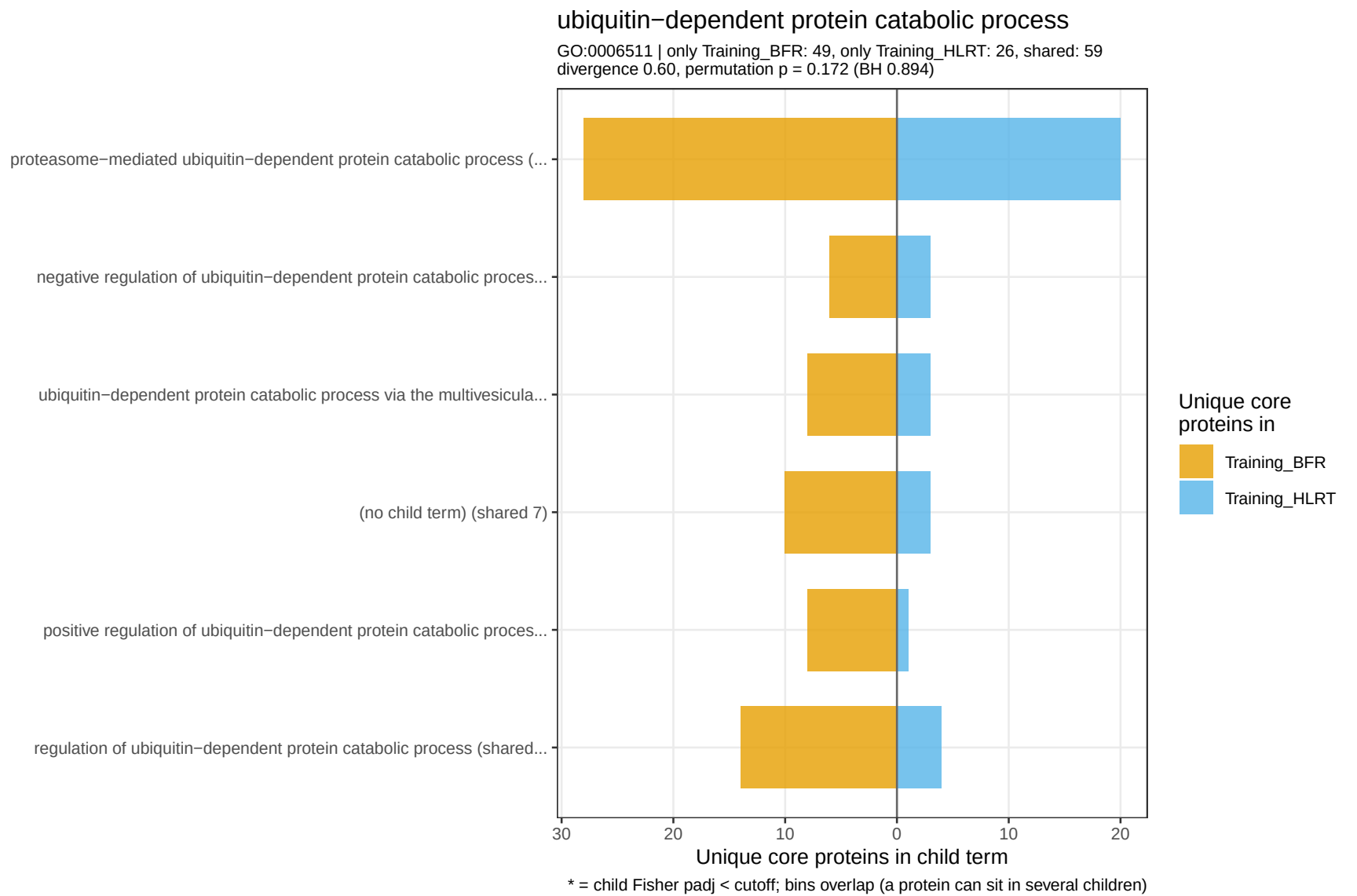

ATP biosynthetic process
GO:0006754 | only Training_BFR: 15, only Training_HLRT: 14, shared: 31
divergence 0.67, permutation p = 0.0885 (BH 0.667)
proton motive force−driven ATP synthesis (shared 31)
(no child term) (shared 0)
positive regulation of ATP biosynthetic process (shared 1)
regulation of ATP biosynthetic process (shared 2)
10
5
0
5
10
15
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

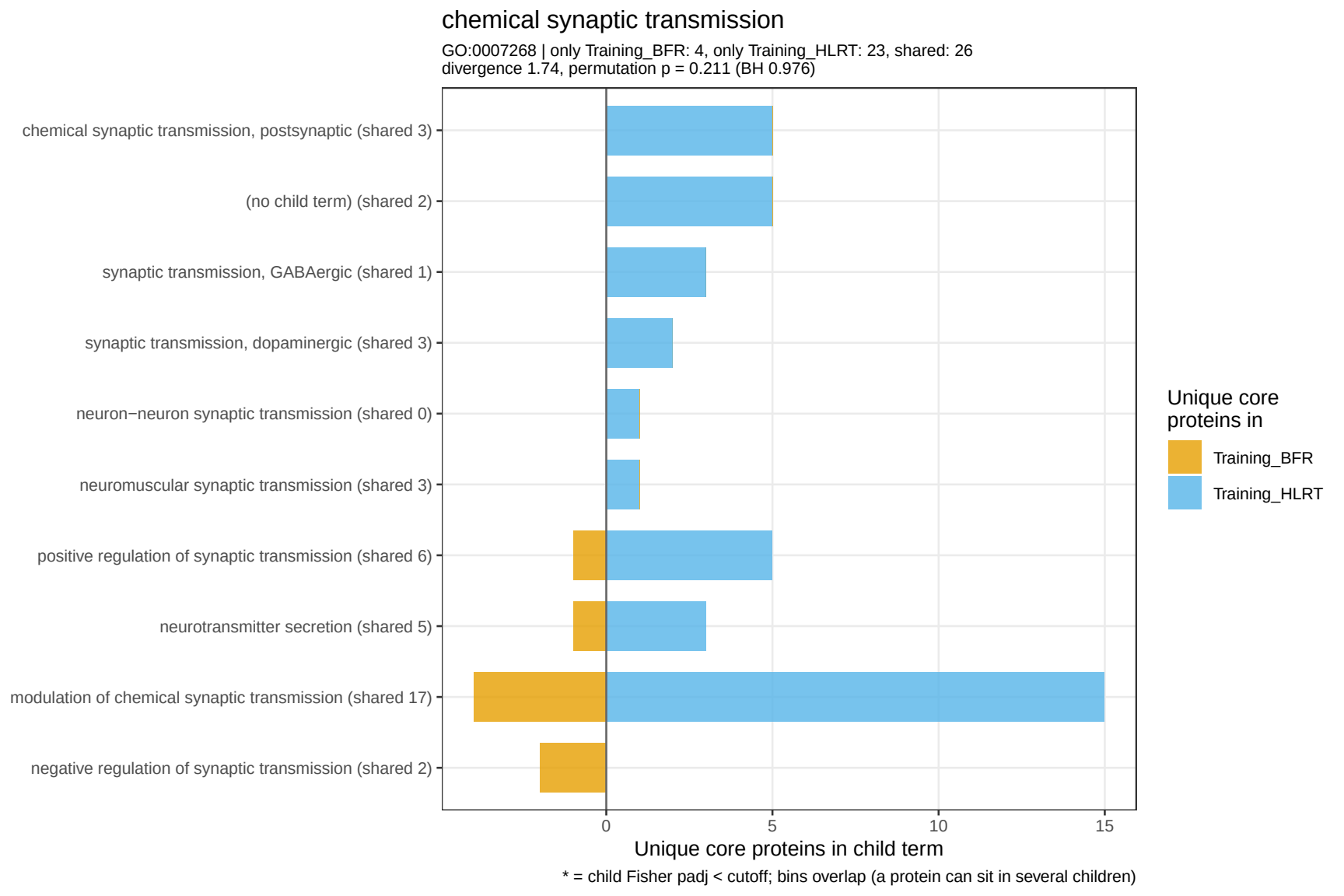

proteolysis
GO:0006508 | only Training_BFR: 54, only Training_HLRT: 44, shared: 73
divergence 0.54, permutation p = 0.112 (BH 0.778)
proteolysis involved in protein catabolic process (shared 53)
negative regulation of proteolysis (shared 6)
self proteolysis (shared 1)
regulation of proteolysis (shared 22)
protein processing (shared 4)
positive regulation of proteolysis (shared 13)
(no child term) (shared 10)
40
20
0
20
40
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

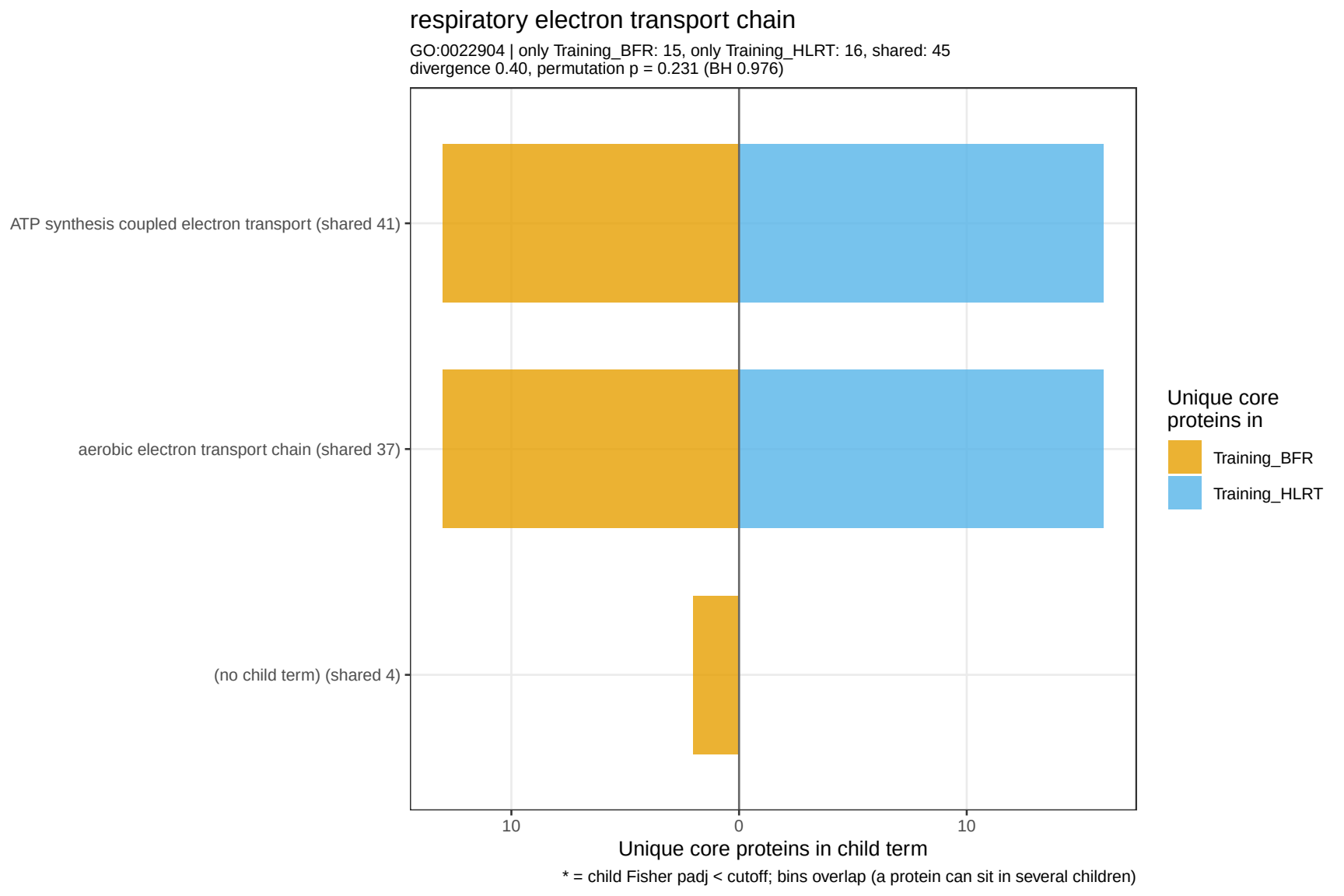

skeletal muscle tissue development
GO:0007519 | only Training_BFR: 5, only Training_HLRT: 3, shared: 24
divergence 1.60, permutation p = 0.148 (BH 0.894)
skeletal muscle fiber development (shared 7)
skeletal muscle cell differentiation (shared 4)
regulation of skeletal muscle tissue development (shared 4)
extraocular skeletal muscle development (shared 2)
(no child term) (shared 10)
1
0
1
2
3
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

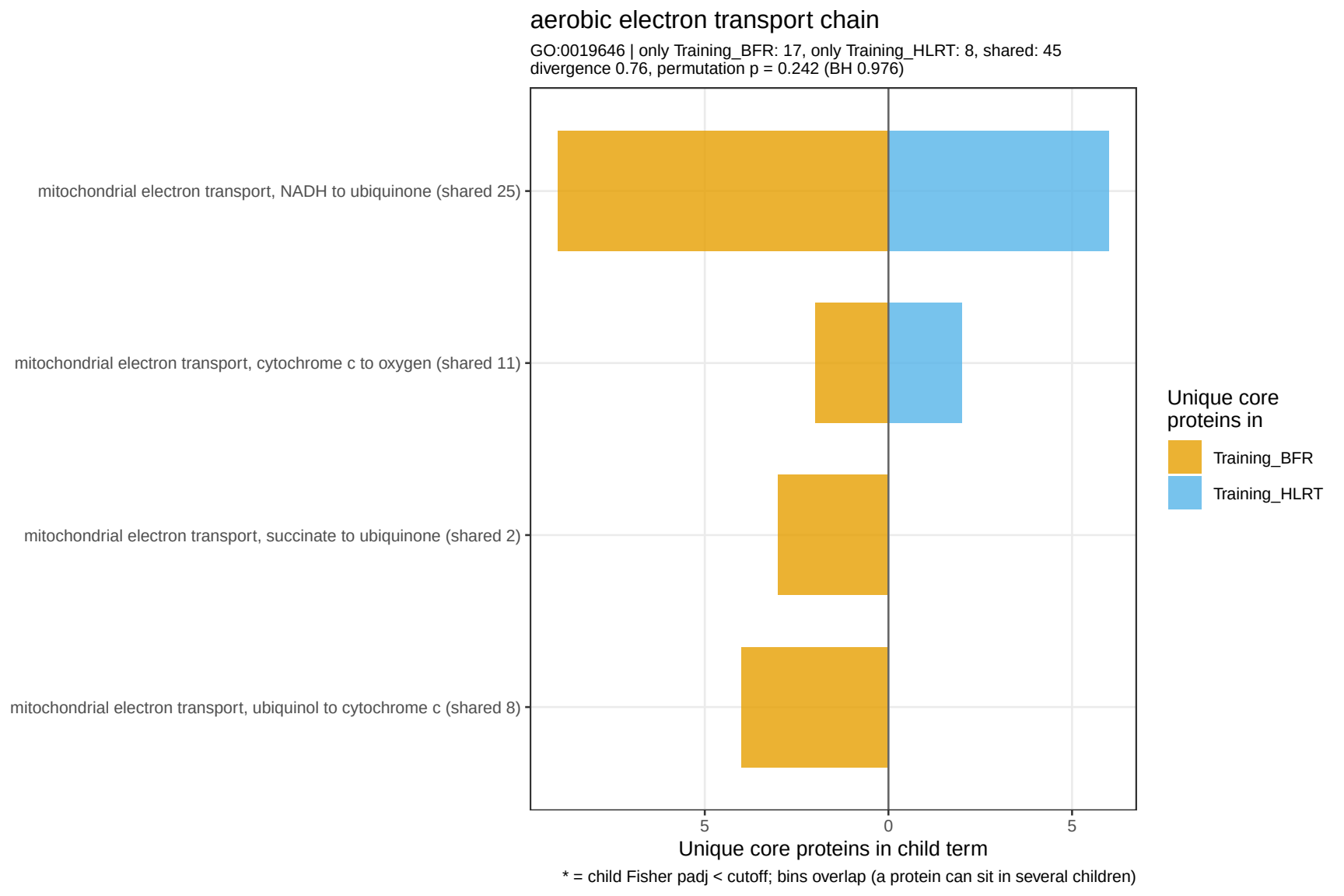

nucleotide biosynthetic process
GO:0009165 | only Training_BFR: 14, only Training_HLRT: 24, shared: 33
divergence 0.73, permutation p = 0.155 (BH 0.894)
ribonucleotide biosynthetic process (shared 30)
purine nucleotide biosynthetic process (shared 33)
pyridine nucleotide biosynthetic process (shared 3)
pyrimidine nucleotide biosynthetic process (shared 1)
nucleotide salvage (shared 4)
positive regulation of nucleotide biosynthetic process (shared 1)
nucleoside monophosphate phosphorylation (shared 1)
deoxyribonucleotide biosynthetic process (shared 0)
(no child term) (shared 0)
regulation of nucleotide biosynthetic process (shared 2)
10
0
10
20
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

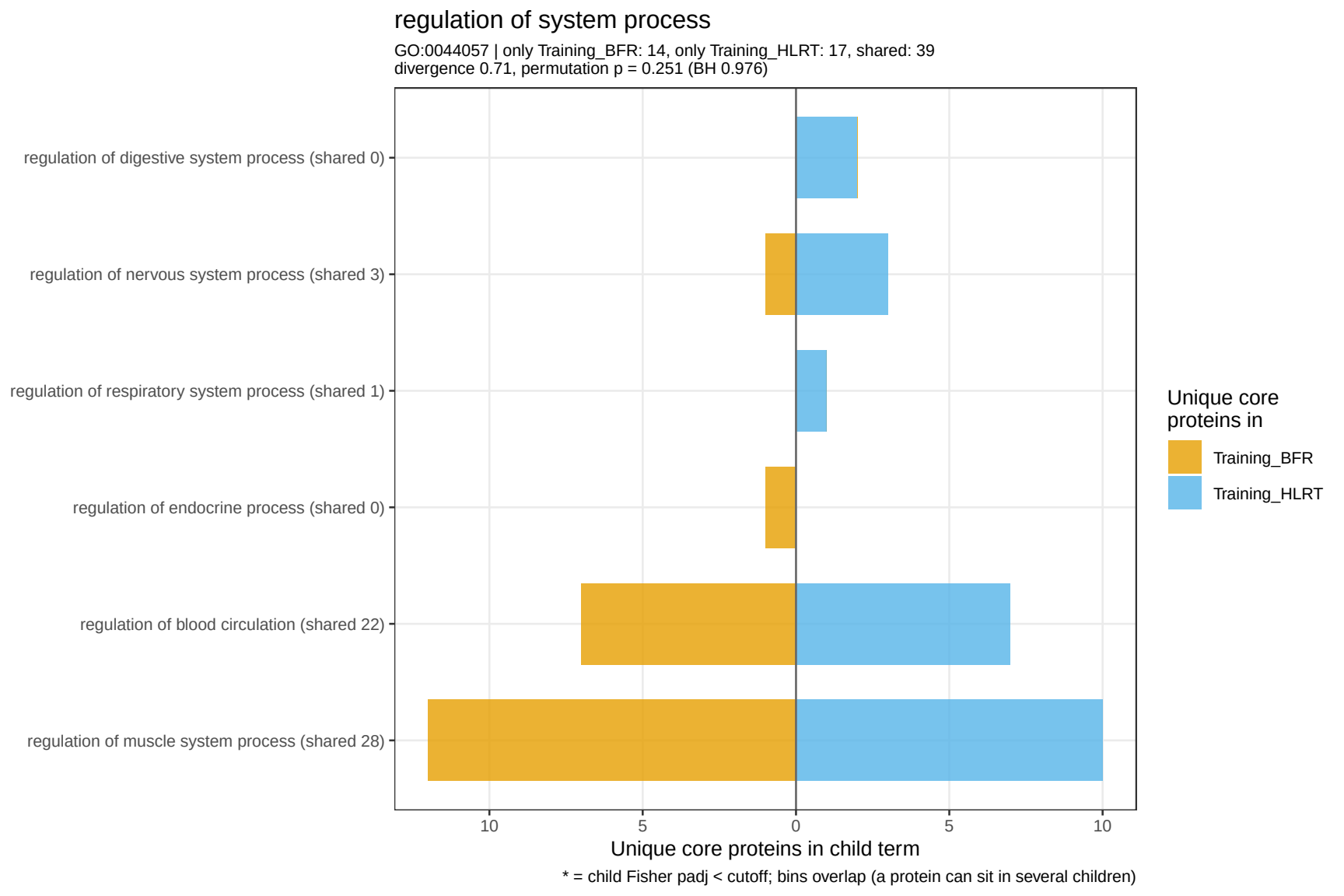

proton transmembrane transport
GO:1902600 | only Training_BFR: 8, only Training_HLRT: 22, shared: 45
divergence 0.41, permutation p = 0.167 (BH 0.894)
(no child term) (shared 42)
energy coupled proton transmembrane transport, against electrochemi...
0
10
20
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

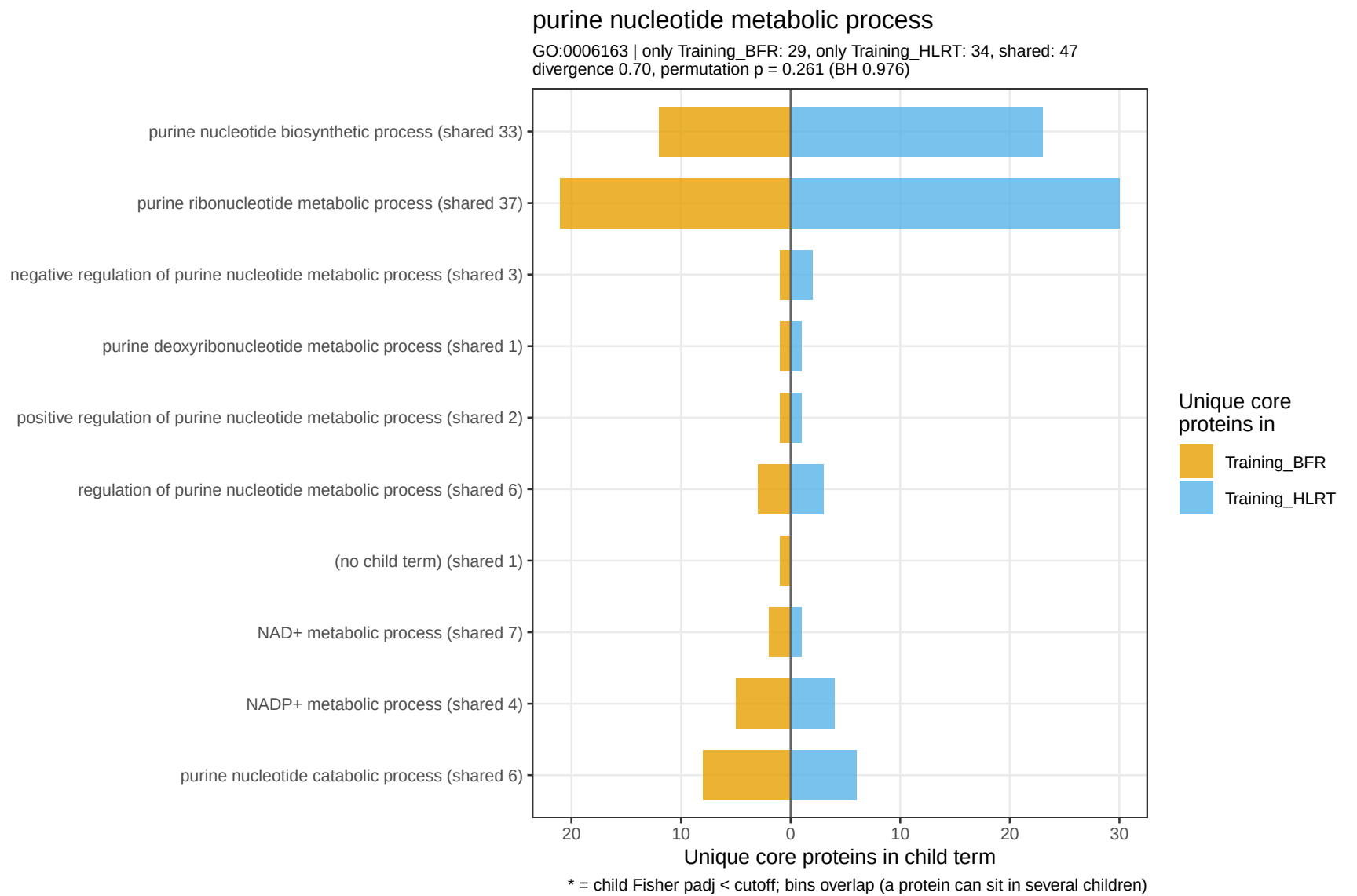

ubiquitin−dependent protein catabolic process
GO:0006511 | only Training_BFR: 49, only Training_HLRT: 26, shared: 59
divergence 0.60, permutation p = 0.172 (BH 0.894)
proteasome−mediated ubiquitin−dependent protein catabolic process (...
negative regulation of ubiquitin−dependent protein catabolic proces...
ubiquitin−dependent protein catabolic process via the multivesicula...
(no child term) (shared 7)
positive regulation of ubiquitin−dependent protein catabolic proces...
regulation of ubiquitin−dependent protein catabolic process (shared...
30
20
10
0
10
20
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

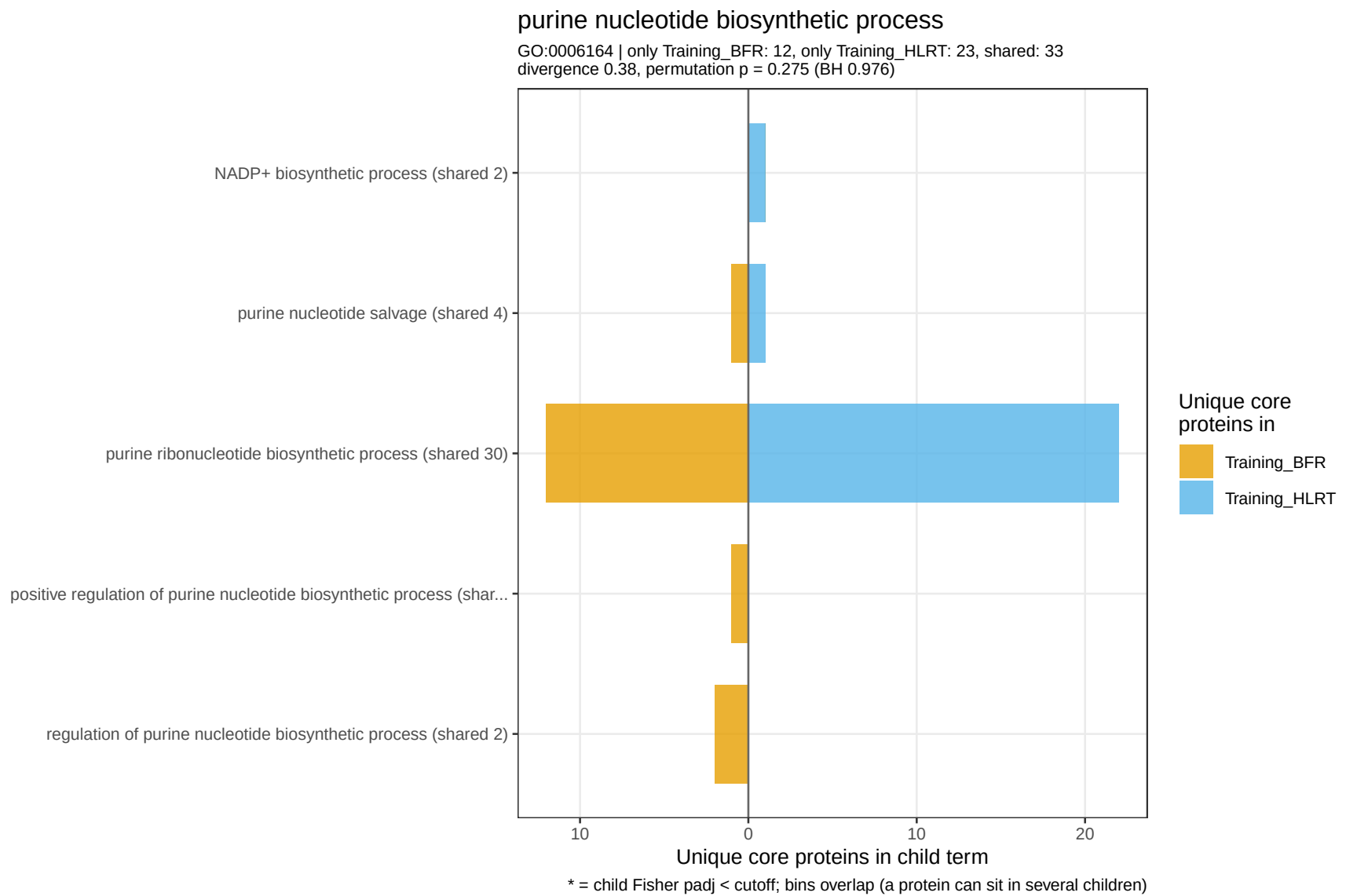

chemical synaptic transmission
GO:0007268 | only Training_BFR: 4, only Training_HLRT: 23, shared: 26
divergence 1.74, permutation p = 0.211 (BH 0.976)
chemical synaptic transmission, postsynaptic (shared 3)
(no child term) (shared 2)
synaptic transmission, GABAergic (shared 1)
synaptic transmission, dopaminergic (shared 3)
neuron−neuron synaptic transmission (shared 0)
neuromuscular synaptic transmission (shared 3)
positive regulation of synaptic transmission (shared 6)
neurotransmitter secretion (shared 5)
modulation of chemical synaptic transmission (shared 17)
negative regulation of synaptic transmission (shared 2)
0
5
10
15
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

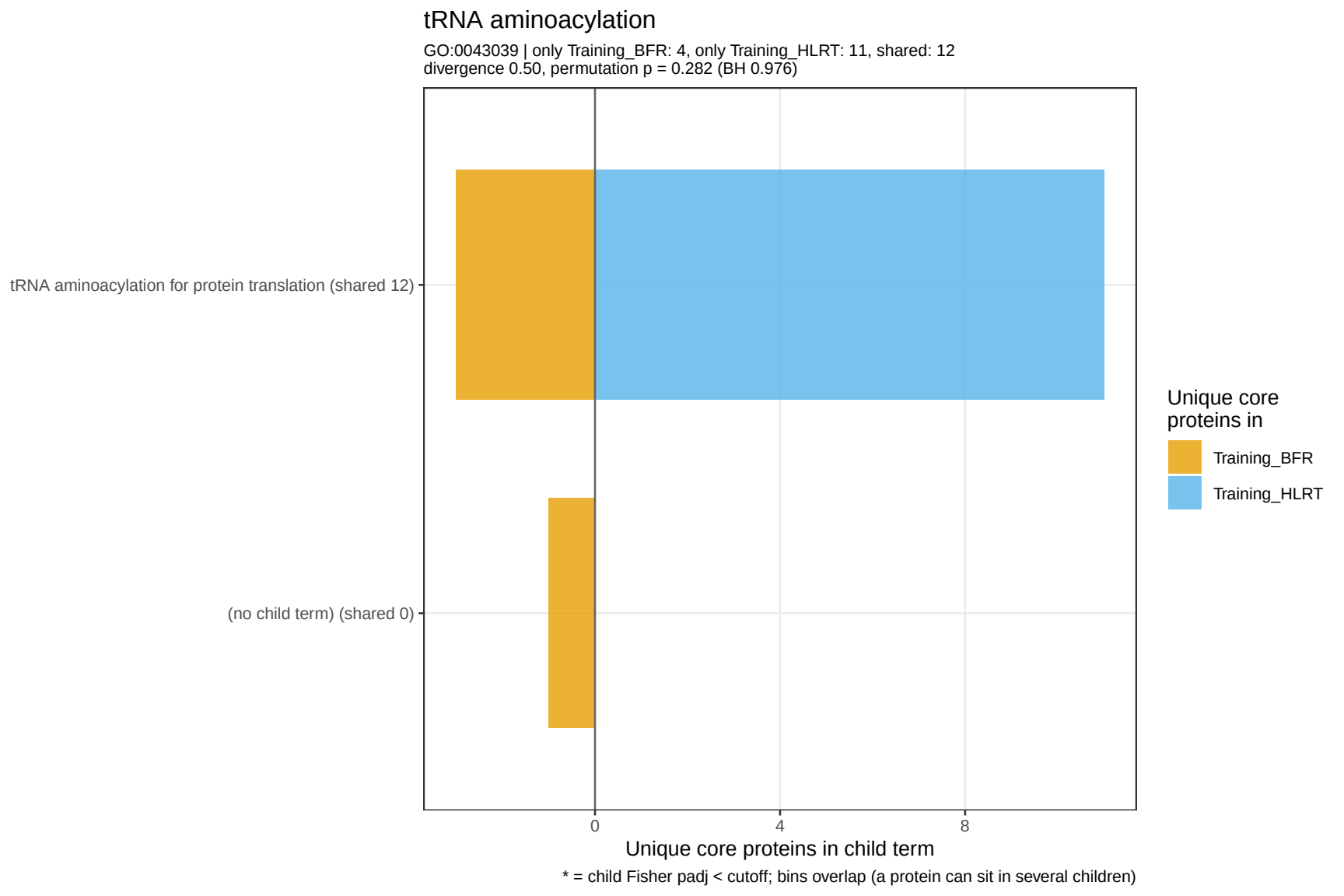

respiratory electron transport chain
GO:0022904 | only Training_BFR: 15, only Training_HLRT: 16, shared: 45
divergence 0.40, permutation p = 0.231 (BH 0.976)
ATP synthesis coupled electron transport (shared 41)
aerobic electron transport chain (shared 37)
(no child term) (shared 4)
10
0
10
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT

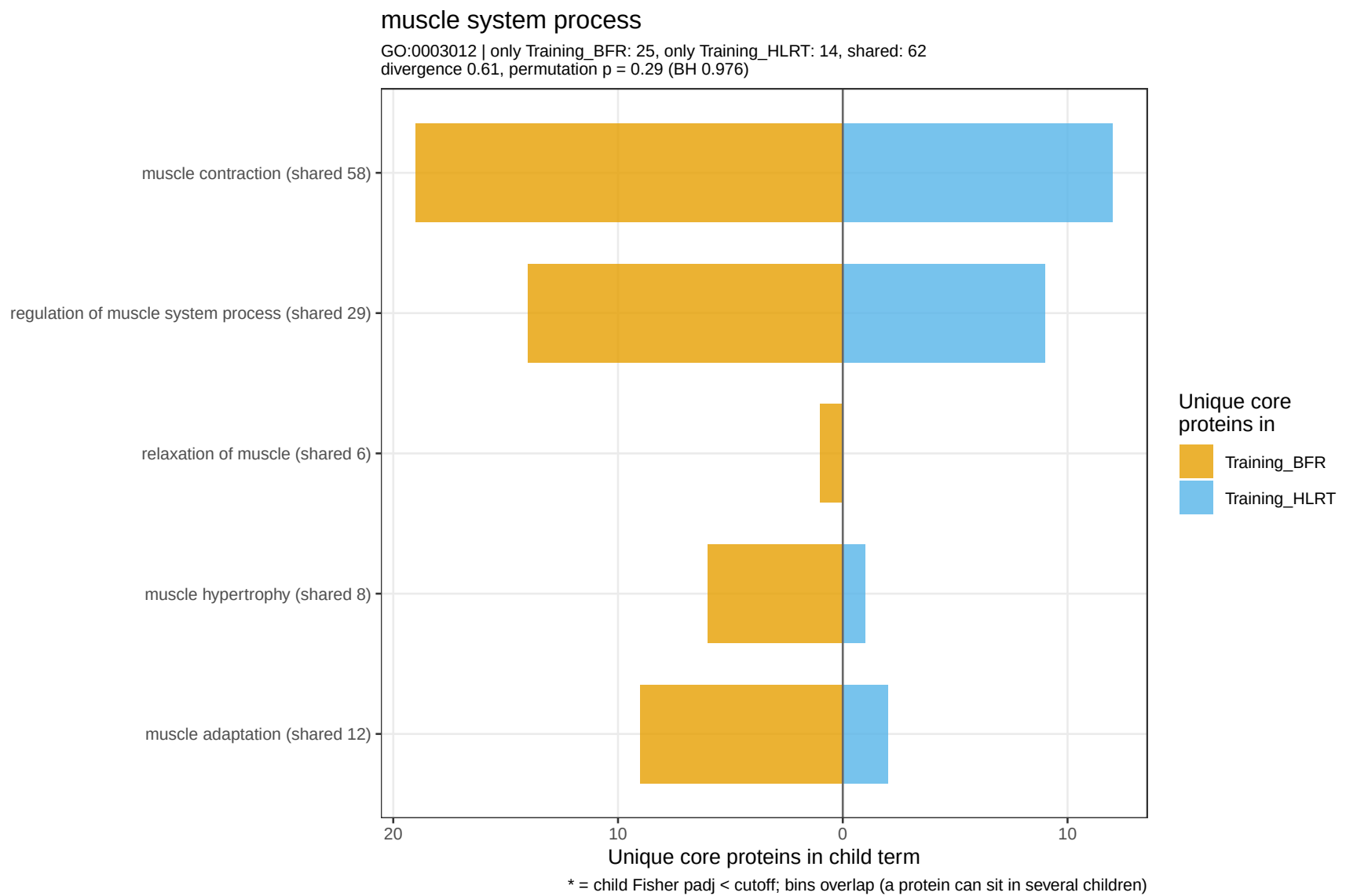

aerobic electron transport chain
GO:0019646 | only Training_BFR: 17, only Training_HLRT: 8, shared: 45
divergence 0.76, permutation p = 0.242 (BH 0.976)
mitochondrial electron transport, NADH to ubiquinone (shared 25)
mitochondrial electron transport, cytochrome c to oxygen (shared 11)
mitochondrial electron transport, succinate to ubiquinone (shared 2)
mitochondrial electron transport, ubiquinol to cytochrome c (shared 8)
5
0
5
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)

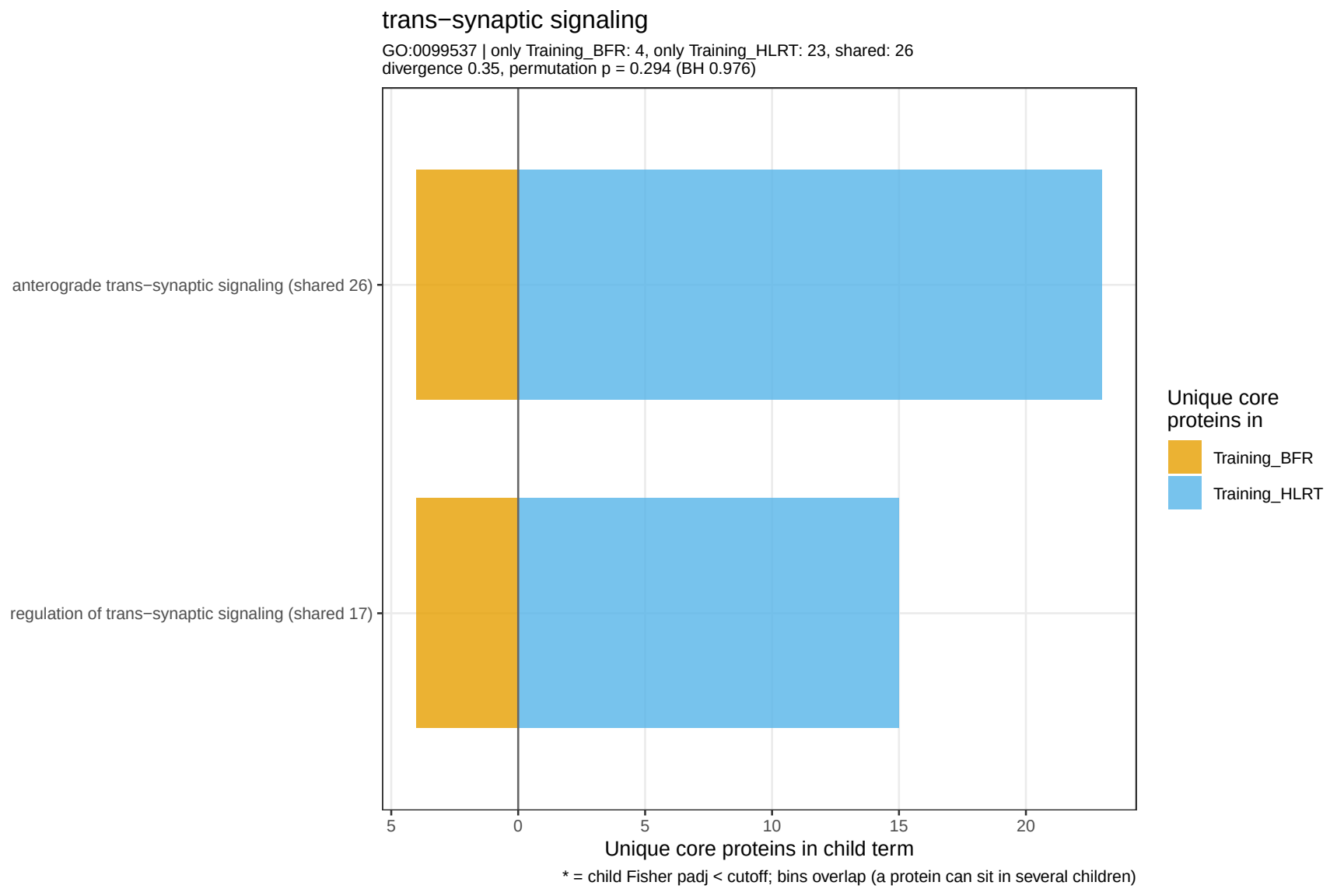

regulation of system process
GO:0044057 | only Training_BFR: 14, only Training_HLRT: 17, shared: 39
divergence 0.71, permutation p = 0.251 (BH 0.976)
regulation of digestive system process (shared 0)
regulation of nervous system process (shared 3)
regulation of respiratory system process (shared 1)
regulation of endocrine process (shared 0)
regulation of blood circulation (shared 22)
regulation of muscle system process (shared 28)
10
5
0
5
10
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT


purine nucleotide metabolic process
GO:0006163 | only Training_BFR: 29, only Training_HLRT: 34, shared: 47
divergence 0.70, permutation p = 0.261 (BH 0.976)
purine nucleotide biosynthetic process (shared 33)
purine ribonucleotide metabolic process (shared 37)
negative regulation of purine nucleotide metabolic process (shared 3)
purine deoxyribonucleotide metabolic process (shared 1)
positive regulation of purine nucleotide metabolic process (shared 2)
regulation of purine nucleotide metabolic process (shared 6)
(no child term) (shared 1)
NAD+ metabolic process (shared 7)
NADP+ metabolic process (shared 4)
purine nucleotide catabolic process (shared 6)
20
10
0
10
20
30
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)


purine nucleotide biosynthetic process
GO:0006164 | only Training_BFR: 12, only Training_HLRT: 23, shared: 33
divergence 0.38, permutation p = 0.275 (BH 0.976)
NADP+ biosynthetic process (shared 2)
purine nucleotide salvage (shared 4)
purine ribonucleotide biosynthetic process (shared 30)
positive regulation of purine nucleotide biosynthetic process (shar...
regulation of purine nucleotide biosynthetic process (shared 2)
10
0
10
20
Unique core proteins in child term
Unique core proteins in
Training_BFR
Training_HLRT
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)


tRNA aminoacylation
GO:0043039 | only Training_BFR: 4, only Training_HLRT: 11, shared: 12
divergence 0.50, permutation p = 0.282 (BH 0.976)
tRNA aminoacylation for protein translation (shared 12)
(no child term) (shared 0)
0
4
8
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT


muscle system process
GO:0003012 | only Training_BFR: 25, only Training_HLRT: 14, shared: 62
divergence 0.61, permutation p = 0.29 (BH 0.976)
muscle contraction (shared 58)
regulation of muscle system process (shared 29)
relaxation of muscle (shared 6)
muscle hypertrophy (shared 8)
muscle adaptation (shared 12)
20
10
0
10
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT


trans−synaptic signaling
GO:0099537 | only Training_BFR: 4, only Training_HLRT: 23, shared: 26
divergence 0.35, permutation p = 0.294 (BH 0.976)
anterograde trans−synaptic signaling (shared 26)
regulation of trans−synaptic signaling (shared 17)
5
0
5
10
15
20
Unique core proteins in child term
* = child Fisher padj < cutoff; bins overlap (a protein can sit in several children)
Unique core proteins in
Training_BFR
Training_HLRT